Tredence — IIT Dhanbad

Reconstructed Online Assessment Pattern & Questions

Component	Details
Total Questions	38
Duration	90 minutes
MCQs	36
DSA Coding	1 question — 20 points
SQL Coding	1 question — 20 points
MCQ Weightage	2 or 3 points each

1. Overall OA Pattern

The Tredence OA consisted of **38 questions** to be completed in **90 minutes**: **36 MCQs, 1 DSA coding question, and 1 SQL question**.

There were many different question sets. The **DSA and SQL questions varied substantially between sets**. MCQs had comparatively less strict uniqueness, so some MCQs could overlap between different sets.

The DSA question supported **multiple programming languages**.

2. MCQ Section — 36 Questions

The MCQ section covered a broad range of topics:

- **Quantitative Aptitude**
- **Probability & Statistics**
- **Data Interpretation (DI)**
- **Machine Learning (ML)**
- **Pseudocode / Output-based questions**
- **English / Grammar**
- **Python:** output-based and syntax-based questions

Individual MCQs carried either **2 or 3 points**.

3. DSA / Coding Section — 20 Points

There was **1 DSA coding question**. The problem could differ significantly between question sets.

DSA Question — Set 1: Snack Selection

A boy wants to eat snacks while maintaining a minimum required level of **satisfaction** and staying within a maximum allowed number of **calories**.

Three arrays are given:

- **A[i]** — satisfaction value of the i-th snack
- **B[i]** — calories of the i-th snack
- **C[i]** — cost of the i-th snack

Given **X** as the minimum required satisfaction and **Y** as the maximum allowed calories, the task is to **minimize the total cost of the snacks selected** while satisfying both constraints.

Constraints remembered: N is the input size of the arrays, with N between **1 and 100**. The array values were within the constraints specified in the question.

Approach used: 3-dimensional Dynamic Programming with index, satisfaction, and calories as the state parameters. The solution was accepted.

DSA Question — Set 2: Second Largest Element to the Right

Given an array, for every element find the **second largest element among the elements to its right**. If a second-largest element does not exist, output **-1**.

Example:

```
Input: 2 4 0 9 6
Output: 9 6 6 -1 -1
```

The DSA problem could be completely different in another question set.

4. SQL Section — 20 Points

There was **1 SQL coding question**. SQL questions also varied substantially between different question sets.

SQL Question — Set 1: Ranking Within Departments

A single table was provided containing columns such as **id, department ID, name, and score**.

The task was to **rank employees within each department in descending order of score**.

The key SQL pattern required ranking independently for every department, using a window-function structure based on **PARTITION BY department** and **ORDER BY score DESC**.

SQL Question — Set 2: Shop Area Calculation

Three tables were provided.

Table 1 — Shop Sales

Column
shop_id
monthly_sales
profit_loss

Table 2 — Shop Type

Column
shop_id
type_id

Table 3 — Type Information

Column
type_id
type_name
rent_per_unit_area

The required output contained **shop_id** and **area**, where the area was calculated using the sales/profit-loss and rent-per-unit-area information.

Area calculation:

```
area = (sales – profit/loss) / rent_per_unit_area
```

The result was required to be **ordered by shop_id**.

Key SQL concepts: joining multiple tables, joining through shop_id and type_id, arithmetic calculation, column aliasing, and ORDER BY.

5. Reconstructed OA Summary

Section	Questions	Points / Weight
MCQs	36	2 or 3 points each
DSA	1	20 points
SQL	1	20 points
Total	38	90 minutes

Question-set characteristic: DSA and SQL were particularly variable across sets, while MCQs had comparatively more overlap between candidates.